

FSD2 LAB • EXPERIMENT 01

ExpressJS Routing, HTTP Methods, Middleware

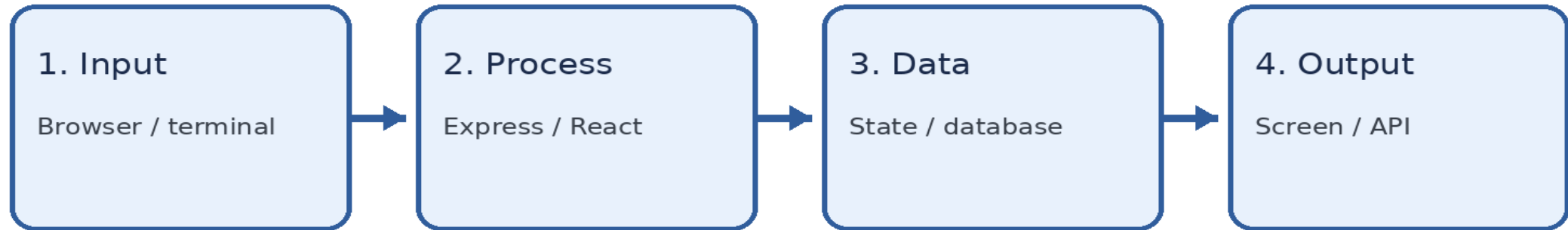
GitHub-readable preview of the presentation. Download the matching .pptx for the editable PowerPoint.

1. Syllabus tasks

- Define a route, handle routes, route parameters, query parameters and URL building.
- Accept data, retrieve data and delete a specified resource using HTTP methods.
- Show the working of middleware.

2. Workflow

ExpressJS Routing, HTTP Methods, Middleware



Workflow: input → processing → data/state → output

3.1. Code pathway — app.js

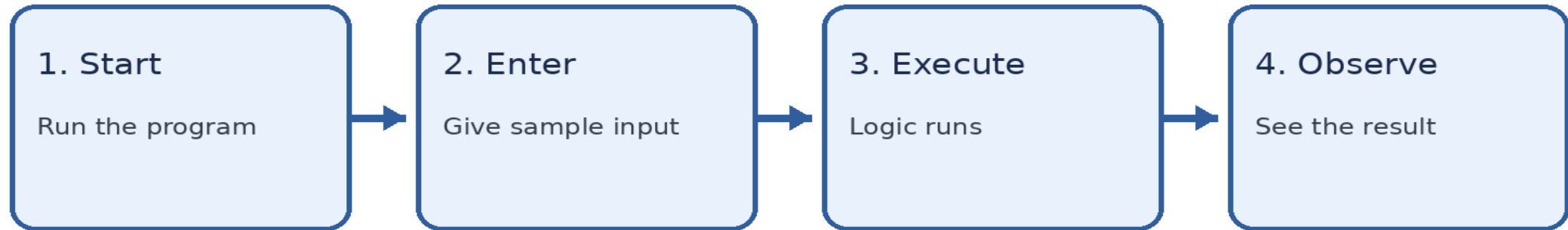
```
1 const express = require('express');
2 const app = express();
3 const PORT = 3001;
4
5 app.use(express.json()); // Read JSON request bodies.
6
7 // Simple custom middleware: runs before the route handler.
8 app.use((req, res, next) => {
9   console.log(req.method, req.url); // Show method and URL in terminal.
10  next(); // Continue to the next middleware/route.
```

3.2. Code pathway — app.js

```
11 });  
12  
13 let students = [{ id: 1, name: 'Rama' }];  
14  
15 app.get('/', (req, res) => {  
16   res.send('FSD2 Express Lab is running');  
17 });  
18  
19 // Route parameter: /students/1  
20 app.get('/students/:id', (req, res) => {
```

4. Runtime walkthrough

Runtime walkthrough — ExpressJS Routing, HTTP Methods, Middlewares



Runtime: start → enter → execute → observe

5. Run and observe

- Start the server / Vite app or open the MongoDB shell as described in the experiment README.
- Enter the sample input or call the required route.
- Watch the terminal for logs and errors.
- Verify the browser, API response, or database result.

6. Viva questions

- What is the purpose of this experiment?
- Which input is required?
- Which line or command performs the main operation?
- What output should you observe?
- What happens when the input changes?